

Pre-Board Examination - 2079

Class : XII

Subject: Physics (1021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The moment of inertia of object does not depend upon
 - mass of object
 - mass distribution
 - angular velocity
 - axis of rotation
- A person standing on a rotating platform has his hands lowered. If he suddenly outstretches his arms, then the angular speed of platform-person system will
 - increase
 - decrease
 - remains constant
 - becomes zero
- 8 rain drops of equal sizes are falling through air with terminal velocity of 10 cm/s. If these drops coalesce, the terminal velocity of big drop is
 - 10 cm/s
 - 20 cm/s
 - 40 cm/s
 - 80 cm/s
- The frequency of vibrating wire is f . When the area of cross-section is halved and the tension is doubled, the frequency becomes
 - f
 - $2f$
 - $3f$
 - $5f$
- The reference intensity of audibility is 10^{-12} Wm^{-2} . The sound level for intensity 10^{-4} Wm^{-2} will be
 - 108 dB
 - 8 dB
 - 80 dB
 - 160dB

6. The specific heat capacity of an ideal gas under isothermal condition is
a. 0 b. 1 c. infinity d. 8.31
7. Assume that a reversible heat engine is operating between source at temperature T_1 and sink at temperature T_2 . If temperature T_2 decreases, the efficiency of heat engine
a. decreases b. remains constant
c. increases d. all of above
8. In an ac circuit, energy is dissipated in
a. resistor only b. inductor only
c. capacitor only d. all resistor, inductor and capacitor
9. Neutral temperature of a thermocouple is 320°C . The cold junction is at 10°C . The inversion temperature is
a. 320°C b. 630°C c. 640°C d. 650°C
10. In a coil of N turns, the value of self-inductance varies as
a. N^0 b. N c. N^2 d. N^3
11. When P-N diode is reverse biased, then
a. No current flows
b. The depletion region is increased
c. The depletion region is reduced
d. The height of potential barrier is reduced

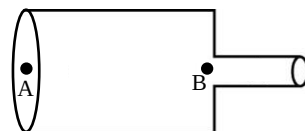
Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. a) Define simple harmonic motion. Derive the expression for the period of oscillation of horizontal mass-spring system. [1+2]
b) A simple pendulum whose length is 1m and oscillates 30 times in one minute at a certain planet. What is the acceleration due to gravity at that planet? [2]

2. a) State Bernoulli principle. Explain why person is advised not to stand near a fast moving train at platform? [1+1]
- b) Water is flowing steadily through the horizontal pipe of non-uniform cross-section as shown in the figure. At point A, the area of cross-section is 48 cm^2 and at point B, the area of cross-section is 12 cm^2 . If the speed of water at B is 4 m/s , then calculate
- The speed of water at A. [1]
 - The pressure difference between points A and B. [1]
 - Volume rate of water flow.
(density of water = 1000 kg m^{-3}) [1]



OR

- a) What is surface tension and discuss the effect of temperature on it? [1+1]
- b) Why does water spread out but mercury collects itself into drops when placed on a clean glass plate? [1]
- c) Calculate the work done in blowing a soap bubble from a radius of 2 cm to 4 cm . The surface tension of the soap solution is $5 \times 10^{-2} \text{ Nm}^{-1}$. [2]
3. Isothermal process is a thermodynamic process in which temperature remains constant.
- On which gas law, isothermal process relies? [1]
 - What could be the change of internal energy during isothermal process? [1]
 - Why adiabatic curves are steeper than isothermal curve? [1]
 - Derive expression for work done in isothermal process. [2]

OR

Petrol engine and diesel engine are four stroke practical engines. They have some fundamental differences in their working.

- What difference is there in the supply procedure of fuel in engine? [1]
- What different process occur during heat supply into petrol and diesel engine? [1]

- c) What are the main reason for higher efficiency of diesel engine? [1]
- d) Draw P-V diagram of both engines. [2]

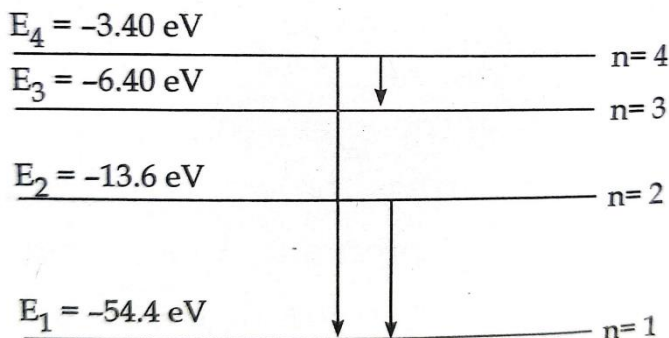
4. a) What is Doppler's effect? Derive an expression for the apparent frequency received by an observer moving towards the stationary source. [1+2]
- b) At what temperature will the velocity of sound in air be twice the velocity at 20°C. [2]

OR

- a) What are the assumptions on which Huygen's principle is based? Does it apply to sound wave in air? [2]
- b) The diffraction of sound wave is more evident in our daily life than light wave. Why? [1]
- c) Derive the equation of a plane progressive wave. [2]
5. a) Distinguish the magnetic properties of diamagnetic, paramagnetic and ferromagnetic materials in terms of permeability and susceptibility. [2]
- b) Why do we prefer steel for making permanent magnet? [1]
- c) The magnetic flux passing perpendicular to the plane of coil is given by $\phi = 4t^2 + 5t + 2$, where ϕ is in Weber and 't' is in second. Calculate the magnitude of instantaneous emf induced in the coil when $t = 2$ sec. [2]
6. a) Discuss the variation of thermo-emf with the variation in temperature of hot junction of a thermo-couple. [3]
- b) The coil of a heater is cut into two equal halves and only one of them is used in heater. What is the ratio of the heat produced by this half coil to that by original coil? [2]

- 7 a) Use Heisenberg Uncertainty Principle to show non existence of electron inside nucleus. [2]

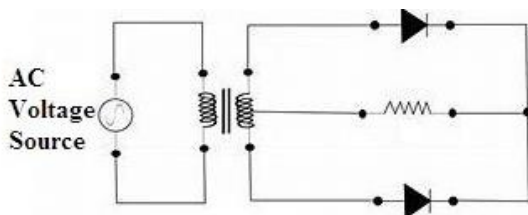
- b) The energy levels of an atom are shown in figure,



- Find the shortest wavelength in the given transitions. [1]
- Find the longest wavelength in given transition. [1]
- Find energy of photon emitted in transition from first excited state to ground state. [1]

8. In the figure, a center tapped full wave rectifier is shown. Answer the following questions based on the figure.

- What do you mean by the term 'rectification'? Name the circuit diagram of given figure. [1]
- What happens if a single diode is used instead of the two in the given circuit? [1]
- How this circuit changes ac into dc? Explain shortly. [2]
- Draw the diagrams for input and output waveforms of voltage. [1]

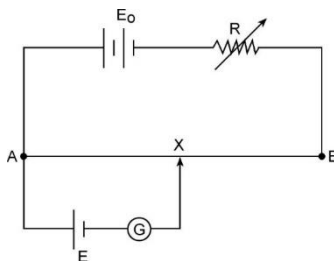


Group - C

Give long answer to the following questions.

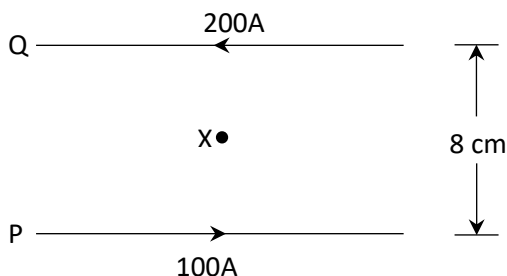
[3 × 8 = 24]

9. a) What are coherent sources? How are they produced? [1+1]
b) Prove analytically that the bright and dark fringes are equally spaced in Young's double slit experiment. [3]
c) A parallel beam of sodium light of wavelength 589.3 nm is incident normally on diffraction grating. The angle between two first order spectra on either side of normal is 28° . Calculate the grating spacing and number of lines per mm on grating surface. [3]
10. a) Explain, how can we convert galvanometer into a voltmeter of given range. [2]
b) A potentiometer wire of length 1m is connected to a driver cell of emf 3V as shown in figure. When a cell of 1.5V is used in secondary circuit, the balance point is found to be 60 cm. On replacing this cell by a cell of unknown emf, the balance point shifts to 80 cm.
i) State principle of potentiometer. [1]
ii) Find potential gradient of the potentiometer wire. [1]
iii) Calculate unknown emf of the cell. [2]
iv) Explain with reason, whether the circuit works, if the driver cell is replaced with a cell of emf 1V. [2]



OR

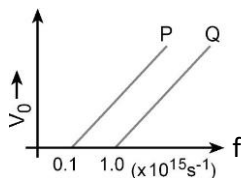
- a) Under what condition, a charged particle moving in magnetic field experiences no magnetic force? [1]
- b) Derive an expression of magnetic field at a point on the axis of a current carrying circular coil. [3]
- c) A long horizontal wire P carries a current of 100 A. It is rigidly fixed. Another wire Q is placed directly above and parallel to P. The wire Q carries a current of 200 A. ($\mu_0 = 4\pi \times 10^{-7} \text{ Hm}^{-1}$)



- i) Find the total magnetic field at midpoint X. [2]
 - ii) Determine the mass per unit length of the wire Q so that it remains suspended due to magnetic repulsion. [2]
11. a) Derive an expression $N = N_0 e^{-\lambda t}$ for a radioactive process, where the symbols carry their standard meanings. [3]
- b) i. Define half-life and decay constant of a radioactive substance. [2]
 - ii. Why Higg's boson is called God particle? [1]
 - iii. The unstable isotope of potassium-40 has a half-life of 2.4×10^8 years. How many decays per sec in a sample containing 2×10^{-6} gm of potassium-40? [2]

OR

- a) Figure shows the variation of stopping potential (V_0) with the frequency (f) of the incident radiation for two photo sensitive metals P and Q.



- i. Explain which metal has smaller threshold wavelength. [1]
 - ii. Explain, giving reason, which metals emits photoelectrons having smaller kinetic energy, for the same wavelength of incident radiation. [1]
- b) How intensity and quality of X-rays can be controlled? [2]
- c) Sir J. J. Thomson devised an experiment for measuring the ratio of charge and mass of electron based on principle of cross field. Describe briefly with diagram an experiment. [3]
- d) An electron is moving in a circular orbit in a magnetic field of $4 \times 10^{-2} \text{T}$. Find time period of revolution [1]
- (Given, $m_e = 9.1 \times 10^{-31} \text{kg}$, $e = 1.6 \times 10^{-19} \text{C}$)

☞ - The End - ☞